

BorneoHack Project Report - Finlytics

Team name : SkippingClasses

1.0 Introduction

1.1 Background of Case Study

Finlytics is an integrated, AI-driven platform designed to modernize the operations of Micro, Small, and Medium Enterprises (MSMEs) across the ASEAN region. These businesses form the economic backbone of the region but remain largely underserved by traditional financial and logistical systems. This project serves as a comprehensive "SME OS," bridging the gap between offline operations and the digital global economy.

1.2 Chosen SDG

SDG 8: Decent Work and Economic Growth – By providing unbanked businesses with access to formal finance and international buyers.

SDG 9: Industry, Innovation, and Infrastructure – Through the creation of AI-driven supply chain hubs and digital identity infrastructure.

1.3 Problem Statement

ASEAN MSMEs face entrenched barriers to formal finance, efficient cross-border trade, and predictable supply chains. The lack of conventional credit histories, real-time demand visibility, trusted digital identities, and automated trade documentation leads to chronic working-capital shortages, missed export opportunities, and high logistics costs.

1.4 Objectives

In our project, we aim to achieve these objectives:

- To qualify unbanked businesses for formal finance using alternative credit scoring.
- To provide hyper-local demand forecasting and smart inventory tracking.
- To automate complex trade regulations and customs documentation via Generative AI.
- To issue verified digital identities that transition offline businesses into formal digital commerce.

2.0 Review of Existing Solutions

2.1 Analysis of Current Solutions

The current landscape is fragmented, with specialized players handling individual pain points:

- **Fintech & Payments:** Players like *Ant Group* and *Grab Financial* offer massive payment data footprints and localized lending but often lack deep integration into supply chain logistics or customs automation.
- **Logistics & Trade:** *Maersk* and *Flexport* provide high trade-lane expertise and digital freight booking, but their services are often high-cost and geared toward larger shippers rather than micro-merchants.
- **Credit Scoring:** *CredoLab* and *Experian* provide sophisticated scoring, but often depend on external partners to provide the actual trade or supply chain context.

2.2 Gaps / Limitations

- **Fragmentation:** No single platform currently bundles credit, supply chain, market analytics, and regulatory automation specifically for the ASEAN MSME context.
- **High Barrier to Entry:** Existing logistics and trade-tech solutions are often too expensive or complex for micro-exporters.
- **Data Silos:** A lack of integrated "provenance" (combining identity, transaction, and shipment data) makes it difficult for lenders to trust unbanked merchants.

3.0 Methodology

3.1 Tech Stack

The platform is built using a modern, full-stack AI-first architecture designed for high performance and scalability.

Frontend

Framework	Next.js 15 and React 19
Styling & UI	Tailwind CSS, paired with Lucide React
Visualizations	Recharts for interactive business analytics, react-simple-maps for geopolitical risk monitor
Content Rendering	React-markdown & remark-gfm

Backend

Framework	FastAPI (Python 3.11+)
AI Orchestration	LangGraph

Protocol	Model Context Protocol via FastMCP
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AI, Data & Services

Core LLM	Google Gemini (gemini-3-flash-preview)
Database & Vector Store	Supabase (PostgreSQL + pgvector)
Memory	Mem0 for persistent conversational memory
Specialized APIs	Tavily, Alpha Vantage, NewsAPI, QuickChart

3.2 Architecture

The platform is built around modular microservices within the project itself, separating AI logic from UI components.

a. Multi-Agent Orchestration Pattern

The system uses a **Supervisor** → **Worker** pattern built on LangGraph:

- **Supervisor (Orchestrator):** Routes user queries to the correct specialized agent based on intent.
- **Researcher Agent:** Equipped with 7 tools for web searching, ASEAN news retrieval, and visual asset generation (charts/images).
- **Quant Agent:** Equipped with 4 tools for financial calculations, stock price tracking, and loan amortization.

b. Functional Service Modules

The backend is organized into domain-specific routers and services, including:

- **Trade Navigator:** Uses RAG (Retrieval-Augmented Generation) to query trade regulation knowledge bases.
- **Credit Engine:** Analyzes 11+ non-traditional data points to generate scores for unbanked businesses.
- **World Monitor:** A real-time command center integrating live USGS earthquake data and global news streams.

3.3 Implementation Process

The implementation followed a phased approach to ensure security and cross-functional integration.

a. AI Agent Hardening

Implementation focused heavily on security, incorporating prompt injection protection, domain-confinement to trade/finance topics only, and input length capping for tool parameters.

b. RAG & Knowledge Integration

- **Semantic Search:** Implemented using Supabase and **pgvector** to allow the AI to "read" and interpret complex ASEAN trade documents.
- **Automated Compliance:** Logic was developed to auto-generate Commercial Invoices and Certificates of Origin based on user inputs and HS code estimation.

c. Full-Stack Deployment Flow

- **Backend Setup:** Configuration via Pydantic settings from **.env** files, with API documentation automatically generated via FastAPI **/docs**.
- **Frontend Integration:** The Next.js frontend communicates with the FastAPI backend through a series of REST endpoints and SSE for AI-driven chat interactions.

4.0 Results

4.1 Summary of Outcome

The project successfully integrated five modular engines into a single dashboard. We've successfully implemented identity verification with transaction history and shipment provenance, the platform can create a "sticky" moat that reduces lender risk and merchant costs.

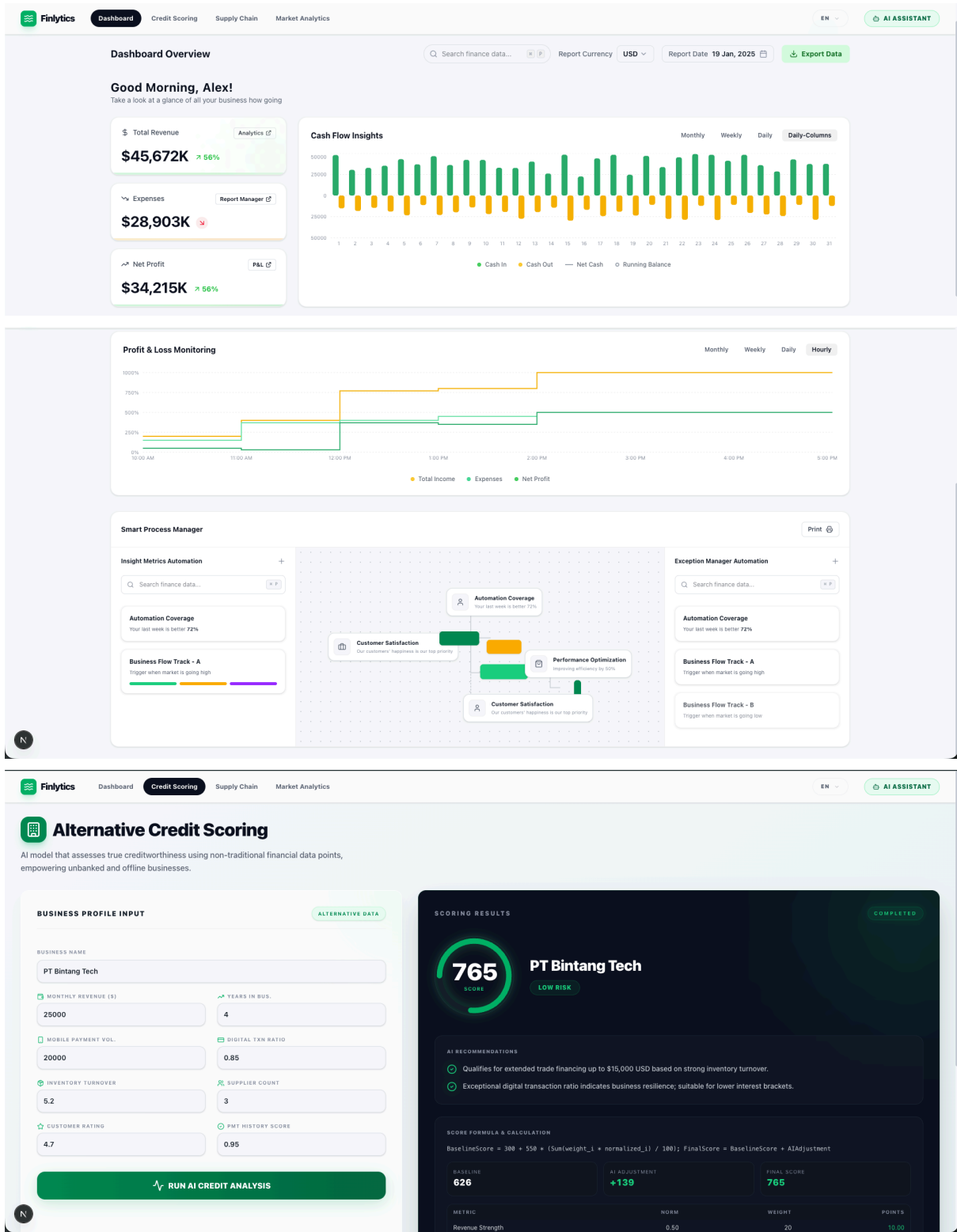
4.2 Data and Feedback

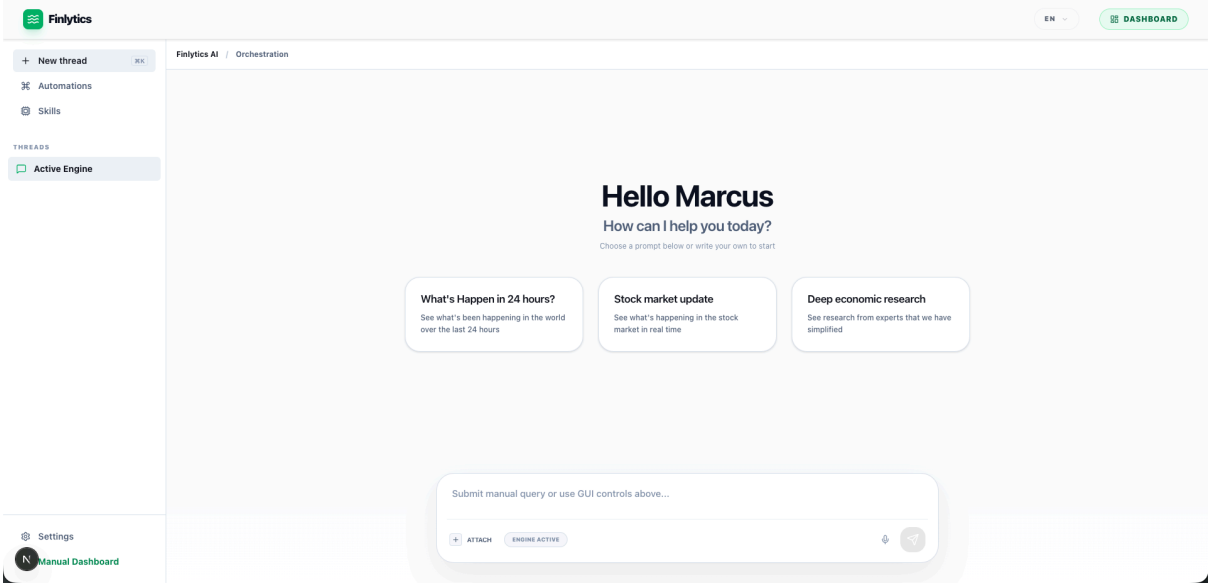
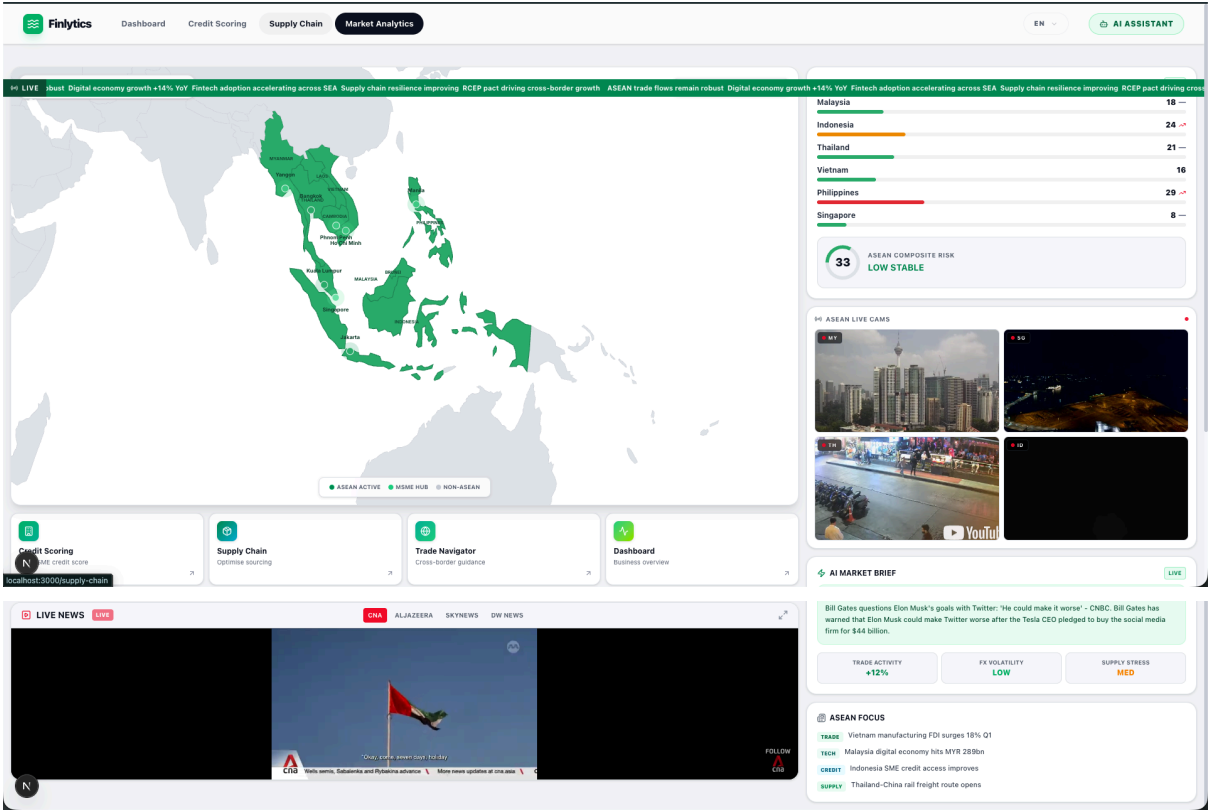
- **Market Capture:** Identified a SAM (Serviceable Available Market) of **USD 7.5 billion** focusing on export-oriented SME segments.
- **Financial Feasibility:** Projected a conservative 5-year revenue (SOM) of **USD 150–400 million** based on pilot corridor data.
- **Validation:** Strategic partnerships with one marketplace and one PSP were identified as the critical "Day 1" requirement to source high-quality ground-truth data.

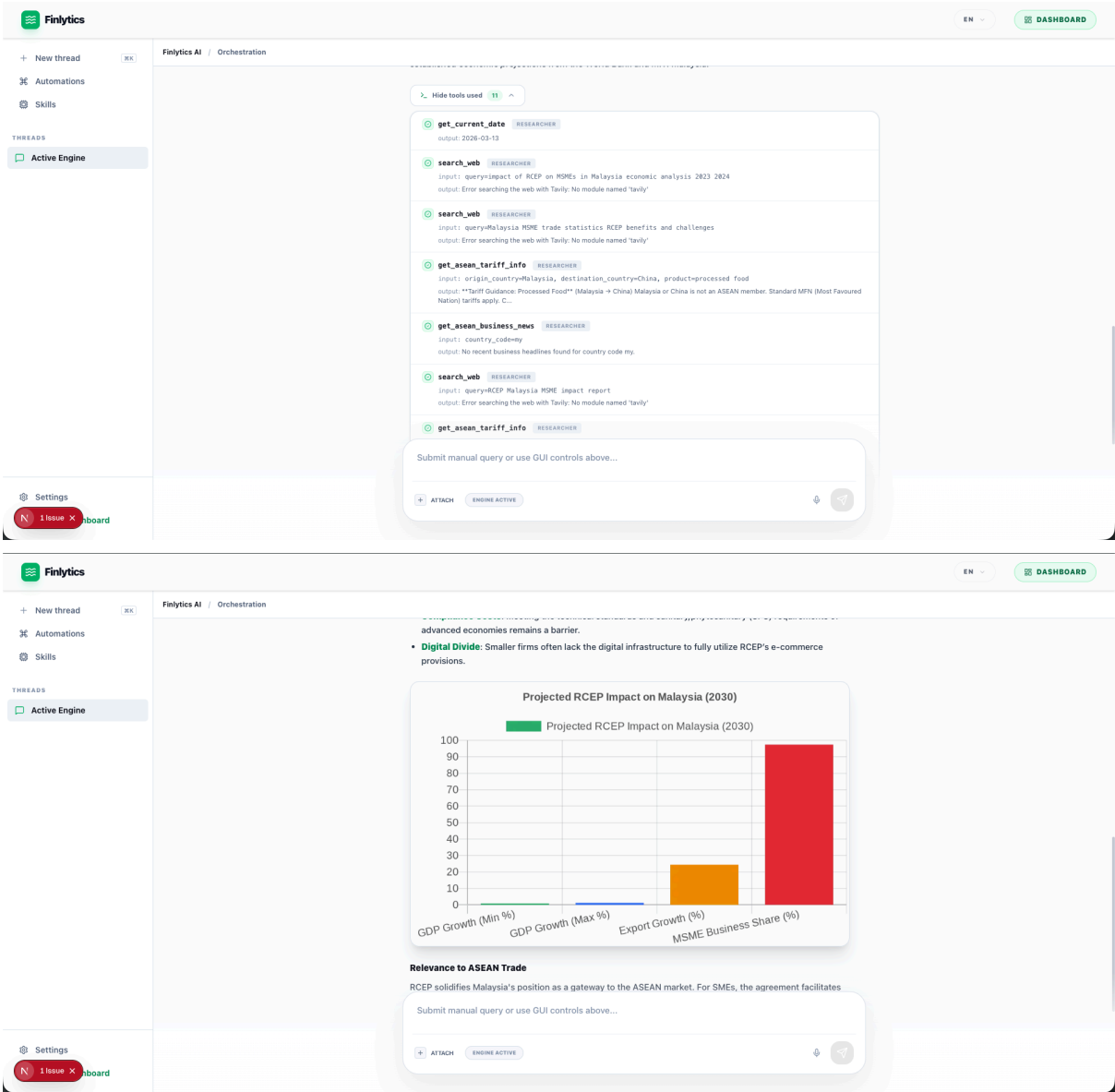
5.0 Prototype Showcase

5.1 Output / UI Screenshots

Include high-quality screenshots of your user interface or key features.







Note: Captions explaining the screenshots are recommended. For example, the caption for  File explains the main navigation flow.

6.0 AI Acknowledgement

6.1 Disclosure & Usage

Use Of AI	Usage Description
Grok / Gemini / ChatGPT	Used for early-stage conceptual validation, comparative analysis of LLM capabilities (leading to the selection of Gemini), and

	generating initial boilerplate code for the FastAPI structure.
VS Code / Codex / Claude Code / Antigravity / Cursor	Utilized for rapid code generation, real-time debugging assistance, and refactoring large components (e.g., separating the LangGraph agents into distinct service files) during the critical implementation phase.

7.0 Conclusion

7.1 Project Summary

Finlytics provides a modular, scalable solution for MSME formalization. By removing the friction of documentation and the barrier of "no credit history," the platform empowers the smallest traders to participate in the regional economy.

7.2 Future Improvements

- **Lender Marketplace:** Transitioning from a platform-led lending model to a competitive marketplace where banks bid to finance MSME shipments.
- **Regulatory Sandboxes:** Working directly with ASEAN governments to integrate the "Merchant Passport" into official national e-KYC systems.
- **Predictive Logistics:** Enhancing the "Hyper-local Demand" engine with satellite imagery and weather data to better assist agricultural exporters.